Republic of the Philippines
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LAB REPORT

(BINARY SEARCH ALGORITHM)

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MACHINE PROBLEM DESCRIPTION:

This program implements and analyzes two essential searching algorithms:

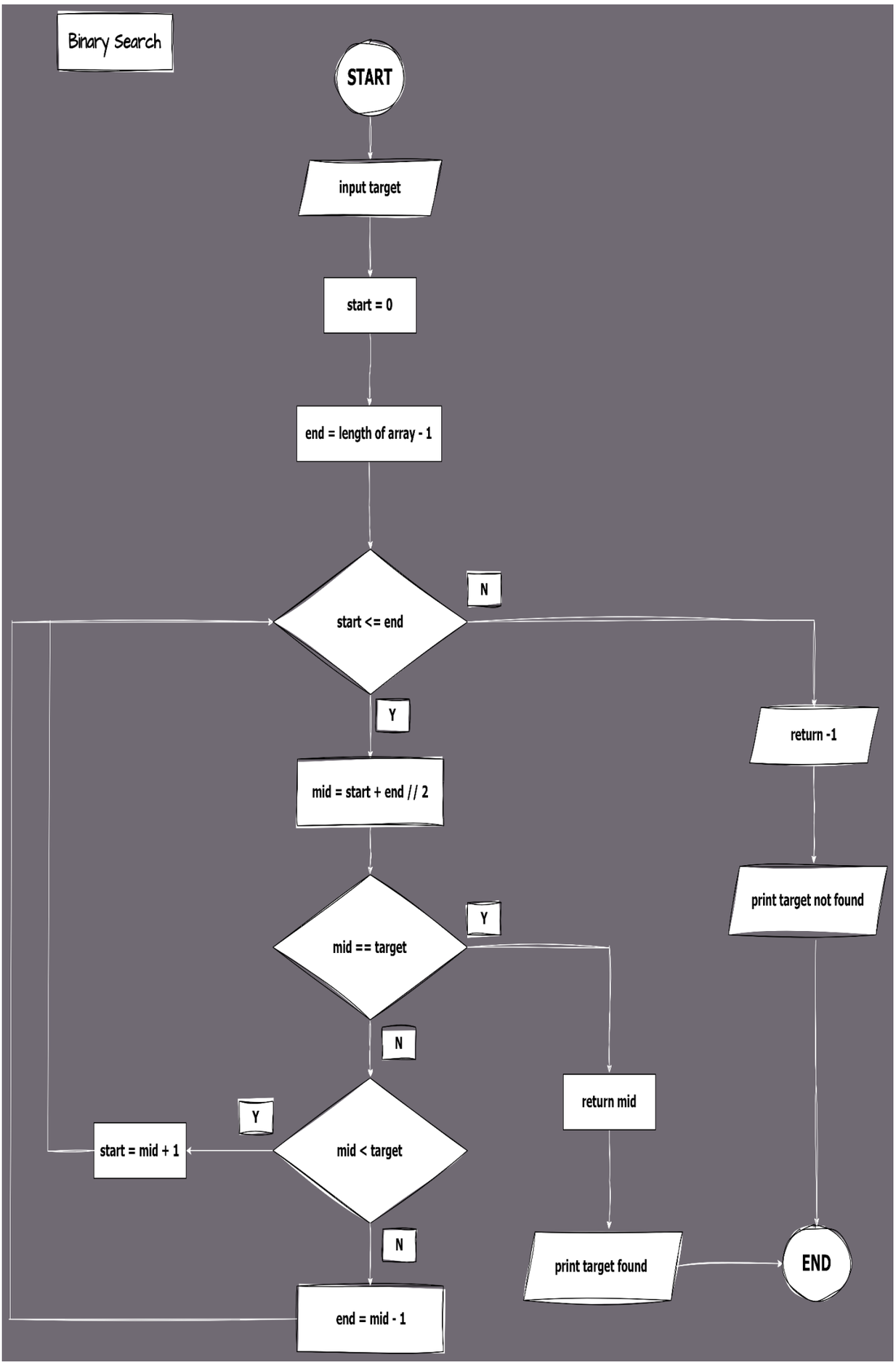
BINARY SEARCH:

This algorithm utilizes a divide-and-conquer strategy, efficiently locating a target element within an ordered dataset by recursively dividing the search interval in half. Its speed and scalability make it ideal for large, organized datasets.

Through this program, we aim to explore the characteristics, advantages, and applications of each algorithm, gaining a deeper understanding of their roles in solving real-world search problems.

FLOWCHART (BINARY SEARCH):

This flowchart represents a scenario where the program processes an array using the Binary Search algorithm. The flowchart details the step-by-step process:



SOURCE CODE:

The code below demonstrates the application of the Binary Search algorithm on an array of integers.

BinarySearch.py

```python
def binary_search(arr, target):
    start, end = 0, len(arr) - 1
    while start <= end:
        mid = (start + end) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            start = mid + 1
        else:
            end = mid - 1
    return -1

if __name__ == "__main__":
    arr = [2, 4, 9, 10, 12, 24, 43, 49, 53, 65, 68, 79, 80, 81, 83, 89]
    target = int(input("Enter the key to search: "))
    result = binary_search(arr, target)
    if result != -1:
        print(f"Element found at index {result}")
    else:
        print("Element not found in the array")
```

OUTPUT:

The output displays whether the target value was found in the array and at which index. If the target value is not present, the output will state, "Element not found in the array."

Element Found:

This output shows the case where the target element is present in the array

```
Run    BinarySearch
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python
Enter the key to search: 49
Element found at index 7

Process finished with exit code 0
```

Element Not Found:

This output shows the case where the target element is not present in the array.

```
Run    BinarySearch
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python
Enter the key to search: 3
Element not found in the array

Process finished with exit code 0
```